

Name: _____ Partner: _____ Period: _____

Context: Modeling Periodic Real-World Data

Sinusoidal functions appear in temperature records, tidal data, daylight hours, and many other natural phenomena. In this activity your group will build and evaluate a sinusoidal model from real-world data. Choose **one tier** based on your readiness. Everyone should be ready to explain the reasoning behind each parameter your group chooses.

Tier R — Reinforce (Key)

Context: The table gives average monthly temperature ($^{\circ}\text{F}$). Max = 80 (month 7), Min = 46 (month 12/Jan area).

Month x	1	2	3	4	5	6	7	8	9	10	11	12
Temp T	48	51	58	65	72	78	80	79	72	63	54	46

Step 1. $A = \frac{80 - 46}{2} = \mathbf{17}$ $D = \frac{80 + 46}{2} = \mathbf{63}$

Step 2. Next max at month **19**, so $k = \mathbf{12}$. $B = \frac{2\pi}{12} = \frac{\pi}{6} \approx \mathbf{0.524}$

Step 3. $C = x_{\max} = \mathbf{7}$

Step 4. $T(x) = \mathbf{17} \cos\left(\frac{\pi}{6}(x - 7)\right) + 63$

Step 5. $T(1) = 17 \cos\left(\frac{\pi}{6}(1 - 7)\right) + 63 = 17 \cos(-\pi) + 63 = 17(-1) + 63 = \mathbf{46}$. Residual: $48 - 46 = \mathbf{2}$; **fairly close.**

Step 6. $T(4.5) \approx 17 \cos\left(\frac{\pi}{6}(4.5 - 7)\right) + 63 = 17 \cos\left(-\frac{5\pi}{12}\right) + 63 \approx 17(0.259) + 63 \approx \mathbf{67.4^{\circ}\text{F}}$

Tier A — Apply (Key)

Context: Tidal height (ft) over 12 hours.

t	0	1	2	3	4	5	6	7	8	9	10	11	12
h	1.5	3.8	5.5	6.2	5.5	3.8	1.5	-0.8	-2.5	-3.2	-2.5	-0.8	1.5

1. Max = **6.2** at $t = 3$ hr. Min = **-3.2** at $t = 9$ hr.

2. $A = \frac{6.2 - (-3.2)}{2} = \mathbf{4.7}$ $D = \frac{6.2 + (-3.2)}{2} = \mathbf{1.5}$

Midline context: **The average tide height above/below mean sea level — approximately where the water would be with no tidal variation.**

3. The pattern repeats every 12 hours (max at $t = 3$, next max at $t = 15$).

$k = \mathbf{12}$ hr $B = \frac{2\pi}{12} = \frac{\pi}{6} \approx 0.524$

4. $h(t) = \mathbf{4.7 \cos\left(\frac{\pi}{6}(t - 3)\right) + 1.5}$

5. $h(0) = 4.7 \cos\left(\frac{\pi}{6}(0 - 3)\right) + 1.5 = 4.7 \cos\left(-\frac{\pi}{2}\right) + 1.5 = 4.7(0) + 1.5 = \mathbf{1.5}$ ft. Residual: **0 (exact match).**

6. Need $h(t) \geq 2$. Setting $4.7 \cos\left(\frac{\pi}{6}(t - 3)\right) + 1.5 = 2$ gives $\cos = 0.106$, so $\frac{\pi}{6}(t - 3) \approx \pm 1.464$, giving $t \approx 3 \pm 2.8$.

Entry is safe approximately on $0 \leq t \lesssim 5.8$ and $0.2 \leq t \lesssim 5.8$ — roughly from $t = 0.2$ hr to $t = 5.8$ hr, and again from $t = 9$ to $t = 12$ hr (when the tide exceeds 2 ft again near the end of the cycle). Students may also express this as approximately hours 0–6 and 9–12.

Teacher Note

Exact computation requires inverse cosine. Accept reasonable approximations read from the data table: tide exceeds 2 ft from about $t = 0$ to $t = 6$ (reading from the table). Tier E students may go further.

Tier E — Extend (Key)

1. City A: max = 75 (mo. 8), min = 58 (mo. 1). $A_A = \frac{75-58}{2} = \mathbf{8.5}$, $D_A = \frac{75+58}{2} = \mathbf{66.5}$, $k_A = 12$, $B_A = \frac{\pi}{6}$, $C_A = 8$. Model: $T_A(x) = 8.5 \cos\left(\frac{\pi}{6}(x - 8)\right) + 66.5$

City B: max = 94 (mo. 7), min = 28 (mo. 1). $A_B = \frac{94-28}{2} = \mathbf{33}$, $D_B = \frac{94+28}{2} = \mathbf{61}$, $k_B = 12$, $B_B = \frac{\pi}{6}$, $C_B = 7$. Model: $T_B(x) = 33 \cos\left(\frac{\pi}{6}(x - 7)\right) + 61$

2. City B has the larger amplitude (**33 vs. 8.5**). **City B experiences far greater seasonal temperature swings — it gets much hotter in summer and much colder in winter than City A, which has a mild coastal climate.**

3. City A: $T_A(1) = 8.5 \cos\left(-\frac{7\pi}{6}\right) + 66.5 \approx 8.5(-0.866) + 66.5 \approx \mathbf{59.1}$. Residual: $58 - 59.1 = \mathbf{-1.1}$.

City B: $T_B(1) = 33 \cos(-\pi) + 61 = 33(-1) + 61 = \mathbf{28}$. Residual: $28 - 28 = \mathbf{0}$ (exact).

4. City B residuals: $T_B(4) = 33 \cos\left(-\frac{\pi}{2}\right) + 61 = 61$; actual 66; residual **+5**. $T_B(7) = 33(1) + 61 = 94$; actual 94; residual **0**. $T_B(10) = 33 \cos(\pi) + 61 = 28$; actual 60; residual **+32**. Large residual at $x = 10$ suggests the sinusoidal model is less accurate for fall months; **the actual fall temperatures stay warmer than the model predicts, perhaps due to thermal lag.**

5. Contextual domain: **$1 \leq x \leq 12$ (one year) for each city.** City B at $x = 14$: $T_B(14) = 33 \cos\left(\frac{\pi}{6}(14 - 7)\right) + 61 = 33 \cos\left(\frac{7\pi}{6}\right) + 61 \approx 33(-0.866) + 61 \approx \mathbf{32.4^\circ\text{F}}$. **This is outside the 12-month contextual domain, so the prediction may be less trustworthy — especially since real temperatures vary year to year and the large residuals in fall suggest the model doesn't perfectly capture City B's seasonal pattern.**